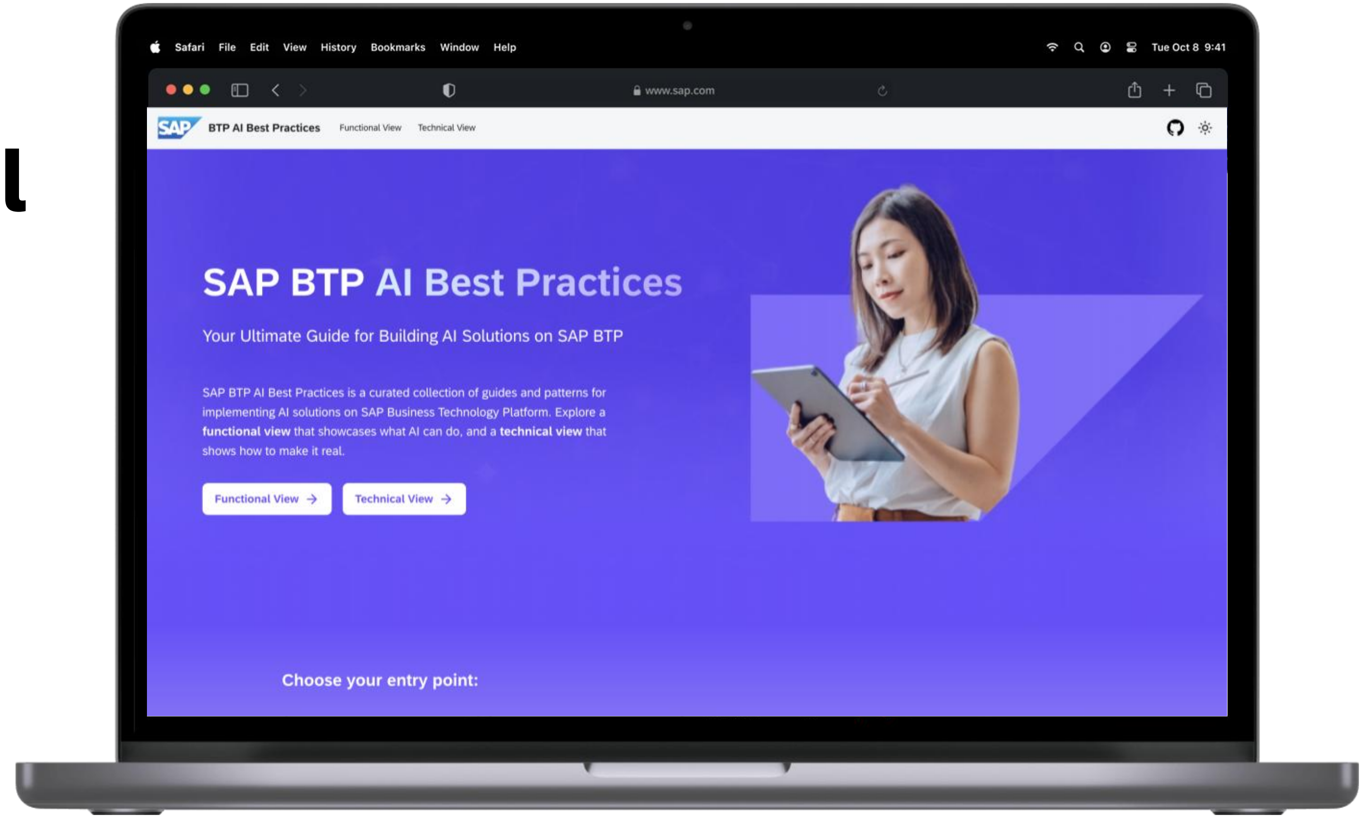
SAP BTP AI Best Practices

Access to SAP RPT Model

A practical guide to using SAP-RPT-1 for zero-training classification and regression on business tabular data.



BTP AI Services Center of Excellence

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Steps

1. Overview
2. Pre-requisites
3. Key Choices and Guidelines
4. Implementation

Access to SAP RPT-1 Model

Table-native predictions without training

SAP-RPT-1 uses **tabular in-context learning**: example rows with known target values teach the task at inference time.

The same request includes query rows where target cells contain **[PREDICT]**, plus target column, task type, index column, and optional schema.

Expected Outcome

- Provide instant, high-quality classification and regression predictions on structured business data without training infrastructure or model fine-tuning cycles.
- Applications can automate tabular prediction tasks while refreshing context rows directly in each request.

Key Benefits

Why use SAP-RPT-1 for tabular prediction?

Zero Training Required

No training pipelines, fine-tuning cycles, or model artifact deployments. Send context examples and predict immediately.

In-Context Adaptation

Representative rows in each request let RPT-1 adapt quickly to the task and current data distribution.

Enterprise Tabular Quality

A table-native model delivers strong classification and regression quality with low maintenance overhead.

Pre-requisites

Business

- SAP AI Core with the “Extended” tier on SAP BTP (Pricing Information)

Technical

- SAP Business Technology Platform (SAP BTP) subaccount (Setup Guide)
- SAP AI Core (Setup Guide)

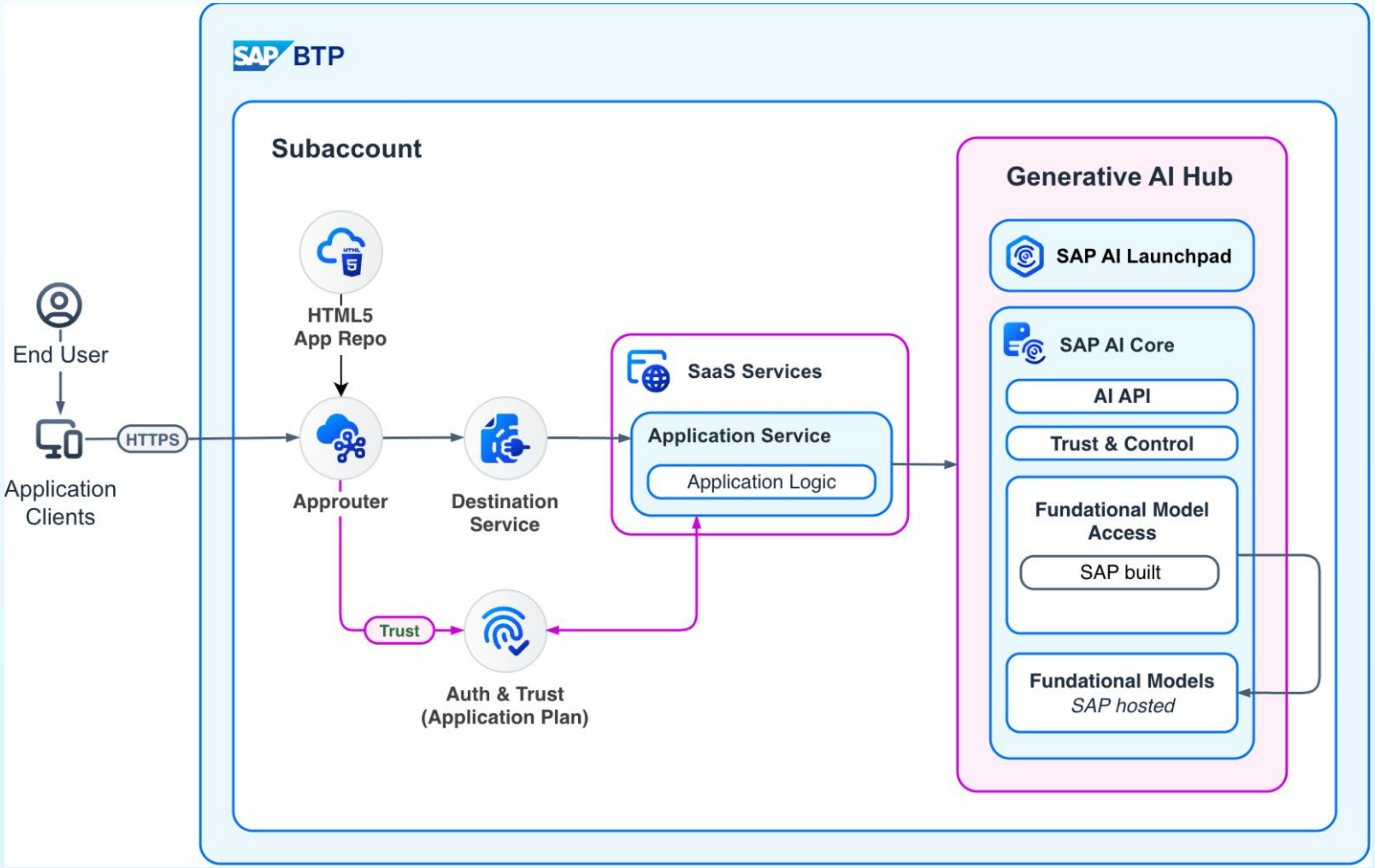
SAP Business Technology Platform (SAP BTP)

- SAP Business Technology Platform (BTP) is an integrated suite of cloud services, databases, AI, and development tools that enable businesses to build, extend, and integrate SAP and non-SAP applications efficiently.

SAP AI Core

- SAP AI Core is a managed AI runtime that enables scalable execution of AI models and pipelines, integrating seamlessly with SAP applications and data on SAP BTP that supports full lifecycle management of AI scenarios.

High-level reference architecture



Key Choices and Guidelines

Model version, context budget, and capacity choices

Model Version Selection
Choose the RPT-1 version based on context capacity, schema width, and latency-quality goals.

sap-rpt-1-small
- Up to **2,048 context rows** and **100 columns**
- Best for **medium-complexity** tasks and **low-latency** throughput
- Start here when the context budget fits

sap-rpt-1-large
- Up to **65,536 context rows** and **256 columns**
- Best for **complex scenarios** and **regression tasks**
- Use when more context materially improves quality

Context Budget Is the Primary Quality Lever

sap-rpt-1-small
Context: 2,048 rows
Columns: 100
Classes: 256 recommended
Use for low latency and medium complexity.

sap-rpt-1-large
Context: 65,536 rows
Columns: 256
Classes: 1,024 recommended
Use for larger context and best quality.

Shared limits: 128 query rows and 10 prediction columns per request. Supported tasks: classification and regression.

Key Choices and Guidelines

Data choices that shape RPT-1 prediction quality and throughput

Data and Request Guidelines

Context selection
- Start with **random** sampling as the baseline
- Use **diversity** sampling for regression
- Use **stratified** or **minority** sampling for imbalanced classification

Batch size
- 16 for debugging or per-row analysis
- 64 as the safe production default
- 128 for throughput when payload size allows

Data quality
- Keep values clean and types consistent
- Provide data_schema and task_type
- Remove leakage columns
- Use meaningful date-derived features

RPT-1 Request Pattern

Context rows Known target values teach the task. > Query rows Target cells contain [PREDICT]. > Predictions Returned with the index column for mapping.

Quality guardrails
Clean data, explicit schema, no target leakage, and feature values available at prediction time.

Operational default
Use batch size 64 first; increase only when payload size and latency allow.

Implementation

SDK paths and shared RPT-1 request concepts

Programming model selection

Choose the SDK that matches your application stack: Python for notebooks and services, JavaScript/TypeScript for Node.js services, and Java for enterprise or Spring Boot applications.

Common implementation pattern

Every language builds the same prediction request: context rows teach the task, query rows mark [PREDICT], and target columns declare classification or regression.

Request Concepts

Context rows: examples with known targets

Query rows: rows with [PREDICT]

Target column: classification or regression

Index column: stable row identifier

Data schema: string, numeric, or date

Python

SDK
- SAP Cloud SDK for AI Python

Reference
- Best Practices Python sample
- SAP-RPT-1 Help

Learning Journeys
- SAP Learning: Introduction to SAP RPT-1

JavaScript/TypeScript

SDK
- @sap-ai-sdk/rpt for Node.js and TypeScript

Reference
- Best Practices TypeScript sample
- SAP AI SDK JS docs

Learning Journeys
- SAP Learning: Introduction to SAP RPT-1

Java

SDK
- com.sap.ai.sdk.foundationmodels:sap-rpt

Reference
- Best Practices Java sample
- SAP AI SDK Java docs

Learning Journeys
- SAP Learning: Introduction to SAP RPT-1

Code Sample

Python

```python
from gen_ai_hub.proxy.native.sap import (
    PredictionConfig,
    RPTClient,
    RPTRequest,
    TargetColumn,
)

client = RPTClient()

body = RPTRequest(
    prediction_config=PredictionConfig(
        target_columns=[
            TargetColumn(name="COSTCENTER", task_type="classification")
        ]
    ),
    rows=[
        {"PRODUCT": "Couch", "PRICE": 999.99, "ID": "35", "COSTCENTER": "[PREDICT]"},
        {"PRODUCT": "Office Chair", "PRICE": 150.80, "ID": "44", "COSTCENTER": "Office Furniture"},
        {"PRODUCT": "Server Rack", "PRICE": 2200.00, "ID": "104", "COSTCENTER": "Data Infrastructure"},
    ],
    index_column="ID",
)

response = client.predict(body=body, model_name="sap-rpt-1-small", model_version="latest")
print(response.predictions)
```

Contributors

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Thank you